Open House Project Expo-2026

Driver Drowsiness Detection System using Web Interface

SDG Goal 3: Good Health , SDG 9 – Industry, Innovation, and Infrastructure ; SDG Goal 11: Sustainable Cities

Team Members

Dr. Bharathi C
Assistant Professor, Dept. of ECE

N/A
8th Semester, ECE

Bhumika KR (1NT22EC036)
8th Semester, ECE

Adithya Krishnan B (1NT22EC007)
8th Semester, ECE

CH Ram Bharadwaj (1NT22EC038)
8th Semester, ECE

Problem Statement:

- **Driver fatigue is a major cause of road accidents**, reducing driver attention, reaction time, and decision-making ability.
- **Lack of proper sleep** due to busy schedules causes fatigue to accumulate over several days.
- **Sudden sleep episodes** can occur when the body becomes extremely tired, leading drivers to fall asleep while driving.
- **Time of day and long working hours** significantly increase the chances of driver drowsiness.
- **Medications and physical health conditions** may also cause sleepiness and reduce alertness.
- **Existing drowsiness detection systems** often require complex computation and expensive hardware.
- **Sensor-based systems such as EEG and ECG** are uncomfortable to wear and not practical for normal driving conditions.

Introduction:

- Driver fatigue is a major cause of road accidents, reducing attention, reaction time, and decision-making ability.
- Lack of proper sleep causes fatigue to accumulate; sudden sleep episodes can occur when the body is extremely tired.
- In India, there is one road accident death every 4 minutes ; over 1.37 lakh people died from road accidents.
- Existing systems require complex computation and expensive hardware (EEG, ECG), making them impractical for common vehicles.
- Sensor-based systems are uncomfortable to wear and not practical for normal driving conditions.
- Night-time accidents often occur due to driver fatigue and loss of vehicle control

e.g., KSRTC Airavat Bus Accident (2023).

Research Survey:

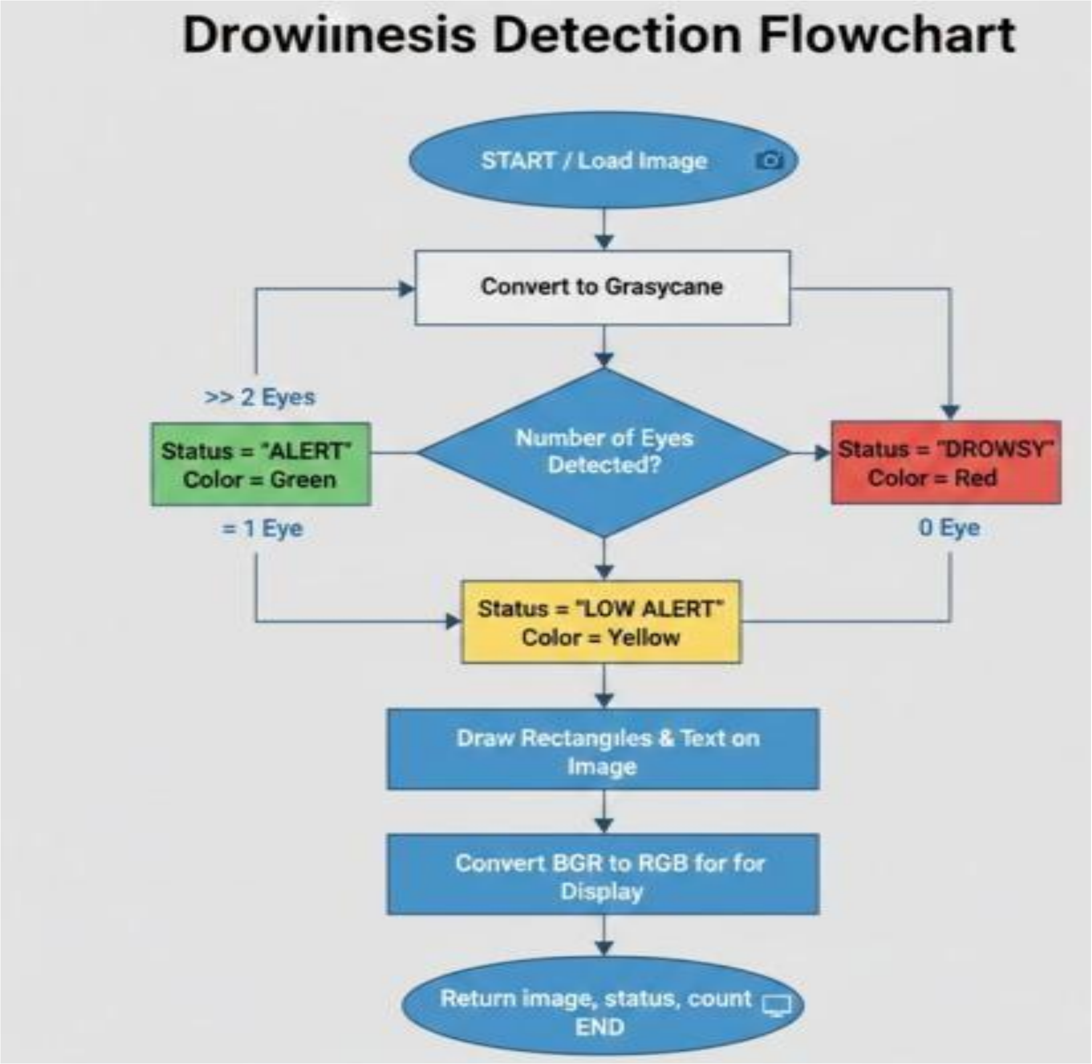
Ref. No.	Authors & Year	Method / Approach	Advantages	Limitations
[1]	S. Abd El-Nabi et al. (2025)	Swin Transformer + Diffusion (image denoising)	Robust under poor lighting, high detection accuracy	High computational cost, not suitable for embedded systems
[2]	D.-L. Nguyen et al. (2024)	Lightweight CNN (eye monitoring)	Real-time performance, low complexity	Focuses only on eye features, ignores yawning
[3]	M. A. Khan et al. (2023)	IoT + Computer Vision (cloud-based)	High accuracy, remote monitoring capability	Requires constant internet, privacy & cost concerns
[4]	S. V. Dixith et al. (2025)	Deep Learning (eye + mouth features)	Improved reliability, multi-feature detection	Higher computational demand
[5]	N. Kumar et al. (2022)	Deep Learning-based real-time system	Generalizable, real-time detection possible	May lack optimization for low-power devices
[6]	S. K. Mishra & M. G. Gupta (2023)	Facial landmarks + CNN	Works well in controlled environments	Struggles with head pose/orientation changes
[7]	M. Venkateswarlu & V. R. R. Ch. (2024)	Lightweight CNN (eye region)	Low complexity, suitable for embedded use	Sensitive to lighting and occlusion
[8]	M. Ramzan et al. (2024)	Hybrid Deep Learning + Feature Extraction	Robust to environmental variations	Complex pipeline, less suitable for low-power systems
[9]	J. Lee et al. (2024)	3D CNN (video-based temporal analysis)	Captures temporal behavior patterns effectively	High resource requirement, difficult for real-time embedded use
[11]	T. Soukupová & J. Čech (2016)	Eye Aspect Ratio (EAR) method	Simple, efficient, widely used for real-time detection	Limited to eye-based detection only
[12]	W. W. Wierwille & F. A. Ellsworth (1994)	Behavioral fatigue analysis	Foundational research, baseline understanding	Not automated, outdated methodology
[13]	C. Lugaresi et al. (2019)	MediaPipe (facial landmark framework)	Fast, efficient, real-time landmark extraction	Requires tuning for edge/low-power devices

Project Objectives

- Build a low-latency, accurate drowsiness detector using camera feed that notifies the driver and remote operators in real-time.
- Develop a responsive web interface for live monitoring, alert logs, and analytics accessible from any device.
- Detect driver drowsiness using Eye Aspect Ratio (EAR) analysis and blink duration monitoring on a continuous video stream.
- Implement a low-cost embedded solution using ESP32-CAM and OpenCV, making it affordable and easy to install in any vehicle.
- Ensure the system operates with minimal latency alert response < 1.5 sec and real-time performance at ~15–20 FPS.

Proposed System Methodology

1. ESP32-CAM captures the driver's face.
2. Video frames are streamed wirelessly via Wi-Fi.
3. Flask server receives and processes the video.
4. Face and eye detection is performed.
5. CNN model classifies eye state.
6. Eye closure duration is monitored.
7. Drowsiness is detected when the threshold is exceeded.
8. Alerts are generated, and notifications are sent.
9. Live status is displayed on the web dashboard



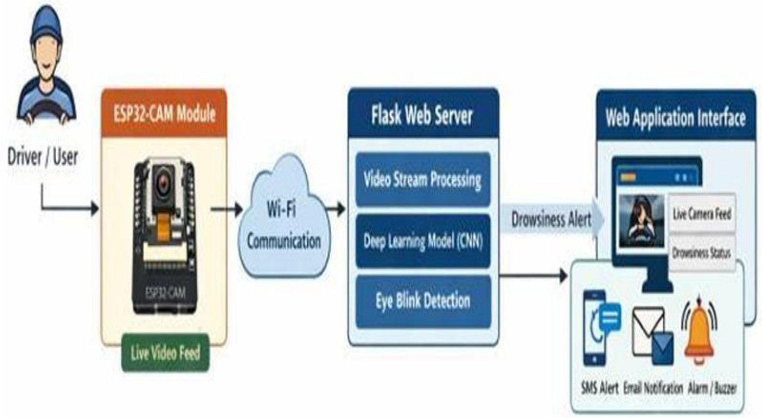
Prototype (if applicable):

Hardware Components:

COMPONENT	DESCRIPTION
ESP-32	Low power SoC Microcontroller
OV2640 Camera	Built-in camera for ESP32
Power Supply	5V supply
Wi-Fi Network	For video streaming
Alert Devices	Buzzer

Software Stack:

SOFTWARE	PURPOSE
Python	Backend programming
Flask + HTML,CSS,JS	Web server & application
OpenCV	Image processing
Operating System	Windows / Linux
Environment	Anaconda Prompt



Prototype (if applicable):

Frontend & Backend Implementation

Frontend (HTML, CSS, JavaScript)

Interactive browser-based dashboard

Displays **live video stream**

Shows real-time metrics: **EAR, MAR, blink rate, driver status**

Dynamic alerts with color indicators:

- Green for Awake
- Red for Eyes Closing / Drowsy

Session event log with timestamps for monitoring

Backend (Flask)

Handles **video processing & state management**

Processes **MJPEG stream frame-by-frame**

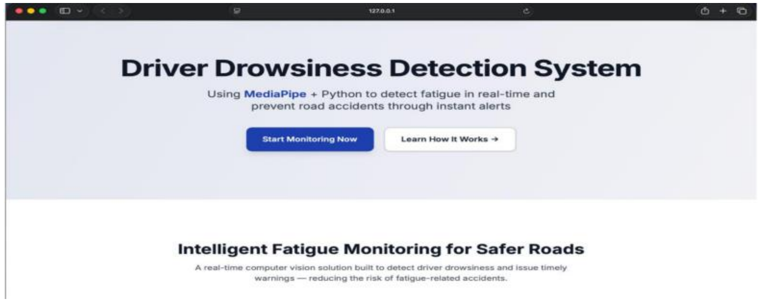
Detects driver condition in real-time

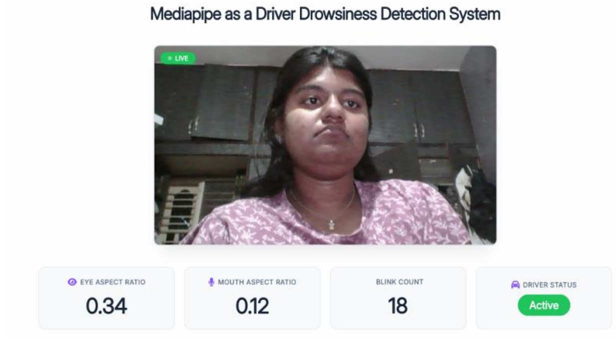
Provides API endpoints:

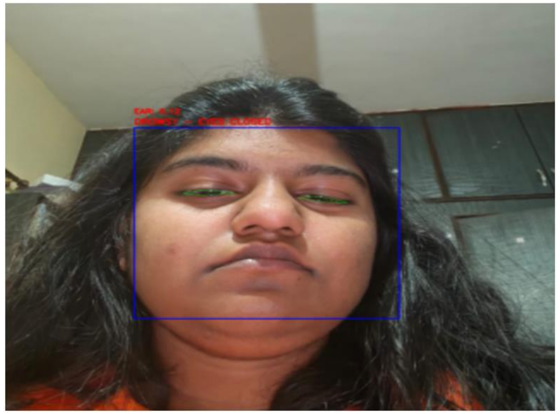
- /video = Live streaming
- /api/state = JSON data (EAR, MAR, status)
- Control = Start / Stop / Reset

System Features

- Real-time monitoring with instant feedback
- Integrated frontend–backend communication
- Continuous logging for analysis
- Lightweight, scalable web-based solution

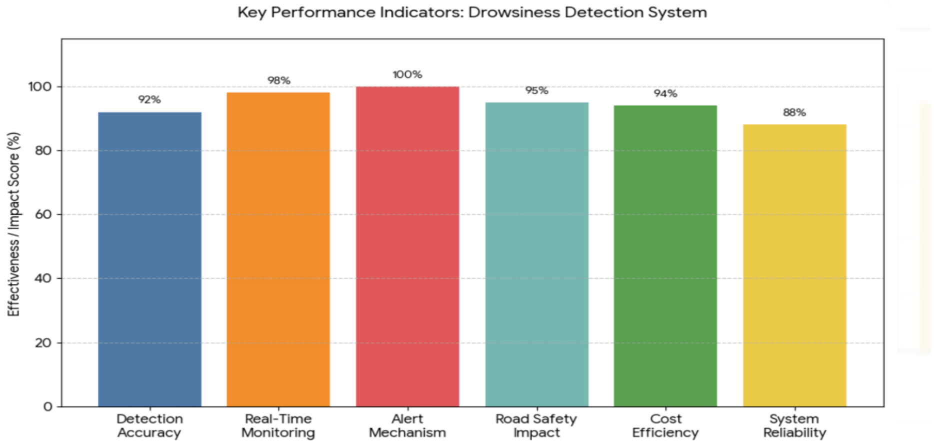


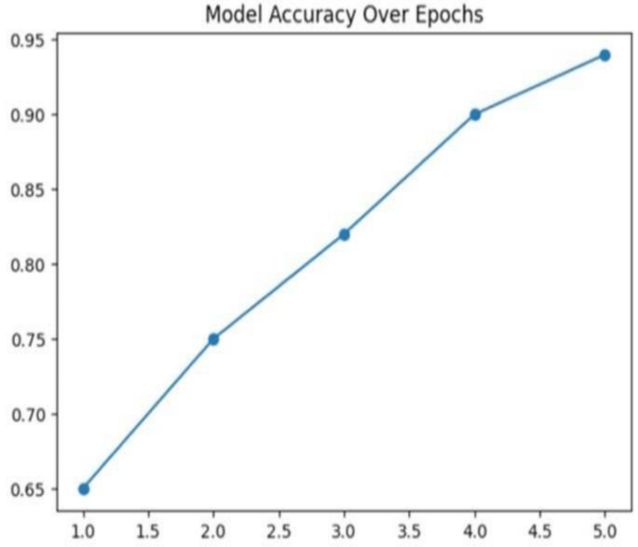




Results and Discussions:

1. **Eye Aspect Ratio (EAR) Threshold:**
   Drowsiness detected when **EAR < 0.25** for **≥ 2 seconds**.
2. **Blink Detection:**
   Normal blink duration **100–400 ms**.
   Drowsiness suspected when **blink duration > 500 ms**.
3. **Blink Rate Monitoring:**
   Normal blink rate **15–20 blinks/minute**.
   Reduced or irregular blinking indicates fatigue.
4. **System Detection Accuracy:**
   Expected **85–95% accuracy** in detecting drowsiness under normal lighting conditions.
5. **Alert Response Time:**
   Buzzer and LED activate within < 1.5 second after drowsiness threshold is crossed.
6. **Processing Speed:**
   Real-time monitoring at approximately **15–20 FPS** using the **ESP32-CAM module**.
7. **Power Consumption:**
   Low power operation **≤ 300 mA at 5V**, suitable for vehicle installation.
8. **System Cost:**
   Estimated prototype cost **₹800 – ₹1500**, making it a **low-cost safety solution**.





Advantages & Disadvantages:

Advantages:

- Low-cost and portable and affordable ESP32-CAM hardware, easy to install in any vehicle.
- Non-intrusive camera-based detection, no wearable sensors; comfortable for drivers.
- Real-time web dashboard for remote monitoring, instant alerts, and historical analytics.
- High expected accuracy (85–95%) using combined EAR, blink rate, and yawning indicators.

Disadvantages:

- Performance may degrade under poor lighting, sunglasses, or extreme head movements.
- Requires stable Wi-Fi connection for web dashboard and remote monitoring features.
- Limited to vision-based cues; does not integrate physiological sensors (EEG/ECG).

Conclusion:

- A real-time, non-intrusive Driver Drowsiness Detection System was developed using ESP32-CAM, OpenCV, Flask, and a responsive web UI.
- The system effectively detects drowsiness through Eye Aspect Ratio (EAR) and blink duration monitoring with expected 85–95% accuracy.
- A responsive web dashboard enables live monitoring, alert notifications, and analytics for driver safety management.
- Prototype cost of ₹800–₹1500 makes it a highly affordable and practical solution for real-world vehicle deployment.
- Future work: improving AI models, adding night-vision, multimodal detection, and cloud-based fleet management integration.
- Addresses SDG Goal 3 (Good Health) SDG 9 – Industry, Innovation, and SDG Goal 11 (Sustainable Cities) by reducing road accidents from driver fatigue.

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Thank You